



DESIGN THINKING

& FABRICATION LABORATORY

About this Course

Design Thinking is a human-centred, iterative approach to innovation that puts *people first* and turns real problems into workable solutions.

- ▶ Blends **empathy**, **creativity** and **rationality**.
- ▶ Moves from *understanding people* to *building and testing* tangible solutions.
- ▶ The **Fabrication Laboratory** turns ideas into physical prototypes using digital fabrication tools.

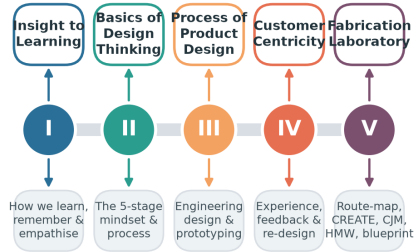
Course structure: five units of 15 hours each, combining theory with hands-on making.

Learning outcomes

By the end of the course you will be able to:

- ▶ Understand how people learn and empathise.
- ▶ Apply the 5-stage design-thinking process.
- ▶ Design, prototype and test products.
- ▶ Map customer journeys and service blueprints.
- ▶ Build solutions in a fabrication lab.

Course Roadmap — Five Units, One Journey



15 hours per unit · 75 hours total · theory + hands-on fabrication

Five units, one continuous journey from insight to fabrication.

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- 1 An Insight to Learning
 - 2 Basics of Design Thinking
 - 3 Process of Product Design
 - 4 Design Thinking & Customer Centricity
 - 5 Fabrication Laboratory



An Insight to Learning

Learning, memory, emotion & empathy

15 hours

Learning is the process of acquiring **knowledge**, **skills** and **attitudes** through experience, study or teaching.

- ▶ **Cognitive** — thinking, understanding, problem solving.
- ▶ **Affective** — emotions, values, motivation.
- ▶ **Psychomotor** — physical / hands-on skills.

Effective designers are *lifelong learners*: they observe, reflect, conceptualise and experiment — exactly the loop Kolb described.

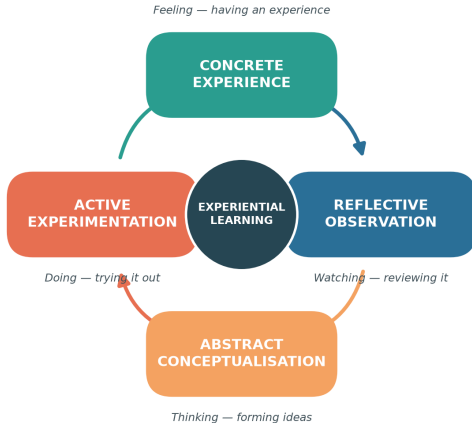
Why it matters for design

Understanding *how humans learn* helps us design products, instructions and experiences that people can pick up quickly and use intuitively.

Kolb's Experiential Learning Cycle

Kolb's Experiential Learning Cycle

Learning as a continuous four-stage loop



Learning is a **continuous cycle** of four stages:

- 1 **Concrete Experience** — do / feel.
- 2 **Reflective Observation** — review.
- 3 **Abstract Conceptualisation** — conclude.
- 4 **Active Experimentation** — plan & try.

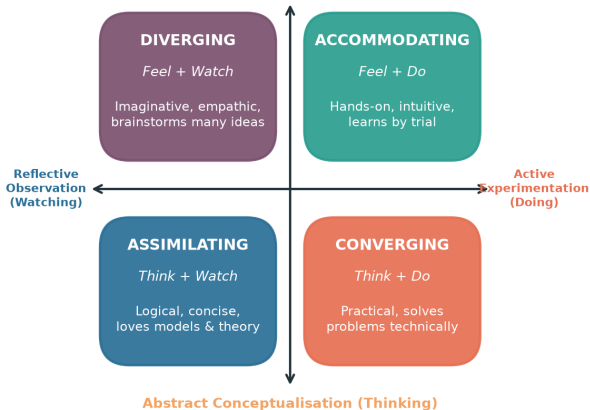
Effective learning happens when we travel *all the way round* the loop — repeatedly.

Kolb's Four Learning Styles

Kolb's Four Learning Styles

Formed by how learners grasp and transform experience

Concrete Experience (Feeling)



Combining *how we grasp* and *how we transform* experience gives four styles:

- ▶ **Diverging** — imaginative, many ideas.
- ▶ **Accommodating** — hands-on, intuitive.
- ▶ **Assimilating** — logical, loves theory.
- ▶ **Converging** — practical problem-solver.

Teams work best when *all four styles* are present.

How we assess

- ▶ **Learning Style Inventory (LSI)** questionnaires.
- ▶ Self-reflection journals.
- ▶ Observation of behaviour in tasks.

How we interpret

- ▶ No style is *better* — each has strengths.
- ▶ Match **teaching methods** to styles.
- ▶ Encourage learners to stretch into weaker modes.

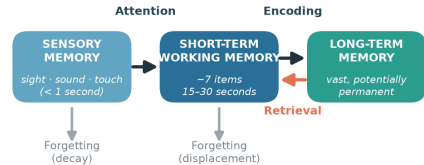
Application with peers

In group projects, deliberately mix styles so ideation, analysis, building and testing are all covered.

The Memory Process

The Memory Process

The information-processing model of memory



Retention problems: interference · weak encoding · lack of cues

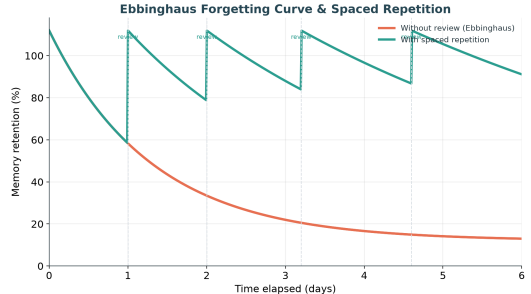
Enhancement: chunking · mnemonics · spaced rehearsal · elaboration

The information-processing model: sensory → short-term → long-term memory.

Problems in Retention

Why do we forget?

- ▶ **Decay** — traces fade without use.
- ▶ **Interference** — old & new memories clash.
- ▶ **Retrieval failure** — no cue to recall.
- ▶ **Weak encoding** — shallow processing.



The **Ebbinghaus curve** shows rapid forgetting — but each review flattens the curve.

Memory Enhancement Techniques

- ▶ **Chunking** — group items into meaningful units.
- ▶ **Mnemonics** — acronyms, rhymes, imagery.
- ▶ **Spaced repetition** — review at intervals.
- ▶ **Elaboration** — connect to prior knowledge.
- ▶ **Dual coding** — words + visuals together.
- ▶ **Active recall** — test yourself, don't re-read.
- ▶ **Method of loci** — memory "palace".
- ▶ **Sleep & rest** — consolidate memories.

Design tip

Good product design *reduces the memory load* on the user — clear labels, consistent layouts and helpful cues.

Emotions shape how people **experience** and **remember** products.

- ▶ **Emotion** — an internal feeling (joy, frustration, trust).
- ▶ **Experience** — the felt journey of using something.
- ▶ **Expression** — how emotion is shown & communicated.

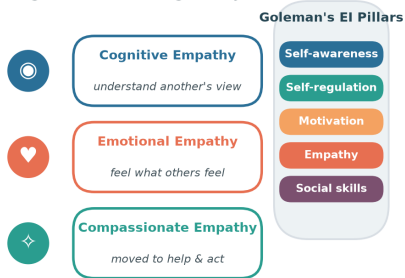
Peak emotions (delight or annoyance) leave the strongest memories — the *peak-end rule*.

Emotional design (Norman)

- ▶ **Visceral** — first look & feel.
- ▶ **Behavioural** — pleasure of use.
- ▶ **Reflective** — meaning & self-image.

Emotions, Empathy & Emotional Intelligence

Understanding emotions is the gateway to human-centred design



Assessing empathy with peers → empathy maps, shadowing, and active-listening interviews

Empathy — the ability to understand and share another's feelings — is the *heart of design thinking*. **Assessing empathy with peers**

- ▶ Empathy maps (says / thinks / does / feels).
- ▶ Shadowing & observation.
- ▶ Active-listening interviews.



Basics of Design Thinking

The mindset, the process, the tools

15 hours

What is Design Thinking?

Definition

A **human-centred**, iterative problem-solving approach that seeks to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

Need for design thinking

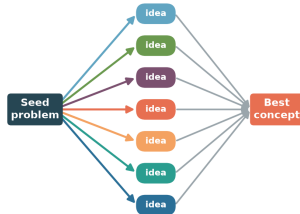
- ▶ Problems are **complex & ill-defined**.
- ▶ Users' real needs are often *hidden*.
- ▶ Reduces the risk of building the wrong thing.

Objectives

- ▶ Keep the **user at the centre**.
- ▶ Encourage **creativity & collaboration**.
- ▶ **Fail early**, learn cheaply.
- ▶ Turn insight into tangible value.

Brainstorming — Divergent then Convergent Thinking

Rules: quantity over quality · build on others · one conversation · stay visual



DIVERGE — generate many ideas, defer judgement, go wild

CONVERGE — group, evaluate & select

Creative work alternates two modes:

- ▶ **Diverge** — generate many ideas, defer judgement.
- ▶ **Converge** — group, evaluate & select.

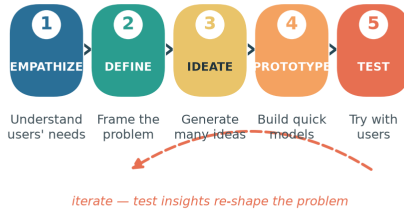
Brainstorming rules

- ▶ Go for quantity.
- ▶ Build on others' ideas.
- ▶ One conversation at a time.
- ▶ Stay visual; welcome wild ideas.

The Five Stages of Design Thinking

The Five Stages of Design Thinking

Non-linear and iterative — teams loop back as they learn



Empathize → Define → Ideate → Prototype → Test — iterative, not linear.

Stages in Detail (1–3)

-
1. **Empathize** Observe and engage with users; conduct interviews and immerse yourself in their world. *Example:* shadowing commuters to learn how they use a ticket machine.
 2. **Define** Synthesise observations into a clear, user-centred **problem statement** (a Point-of-View). *Example:* "Busy commuters need a faster way to buy tickets because queues cause stress."
 3. **Ideate** Generate a wide range of ideas through brainstorming, sketching and mind-mapping — *quantity over quality* first.

Stages in Detail (4–5)

-
- 4. **Prototype** Build quick, inexpensive, scaled-down versions to explore ideas. *Example:* a paper mock-up of the ticket-machine screen.
 - 5. **Test** Try prototypes with real users, gather feedback and refine. Testing often *re-defines the problem* — sending you back to earlier stages.

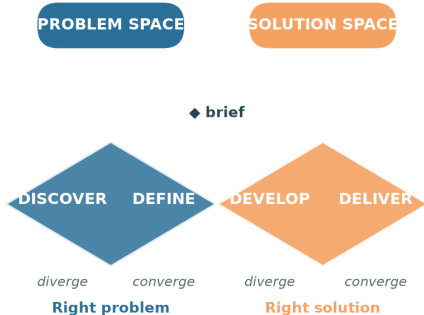
Being ingenious

The stages are a *flexible toolkit*, not a rigid recipe. Teams loop, skip and revisit stages as they learn.

The Double Diamond

The Double Diamond

Alternating divergent (explore) and convergent (focus) thinking

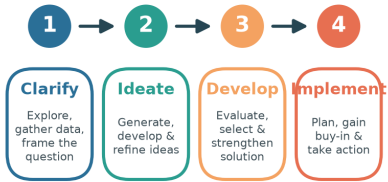


A popular map of the design process:

- ▶ **Discover & Define** — find the *right problem*.
- ▶ **Develop & Deliver** — build the *right solution*.

Each diamond widens (diverge) then narrows (converge).

Creative Problem-Solving Process



Testing creative problem solving: judge originality, fluency, flexibility & elaboration of ideas

Creative problem solving moves through four stages: **Clarify**, **Ideate**, **Develop** and **Implement**. Testing creative problem

solving — judge ideas on:

- ▶ *Fluency* — number of ideas.
- ▶ *Flexibility* — variety of ideas.
- ▶ *Originality* — novelty.
- ▶ *Elaboration* — detail & refinement.

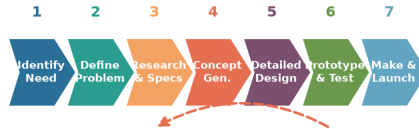


Process of Product Design

Engineering design, prototyping & testing

15 hours

Engineering Product-Design Process



iterate on test results

design-thinking approach keeps the user in the loop at every stage

A structured flow from identifying a need to manufacture & launch.

Design-Thinking Approach to Product Design

Traditional engineering

- ▶ Requirements defined up front.
- ▶ Linear, specification-driven.
- ▶ User feedback comes late.

Design-thinking infused

- ▶ **User needs** continuously explored.
- ▶ **Iterative** — prototype & test early.
- ▶ Feedback loops at every stage.

Best in class

Great products (e.g. intuitive smartphones, ergonomic tools, well-designed appliances) combine *function*, *form* and *delight* — form follows function *and* feeling.

What is a Prototype?

A **prototype** is an early, testable model of a product used to *explore ideas* and *learn* before committing to production. **Why prototype?**

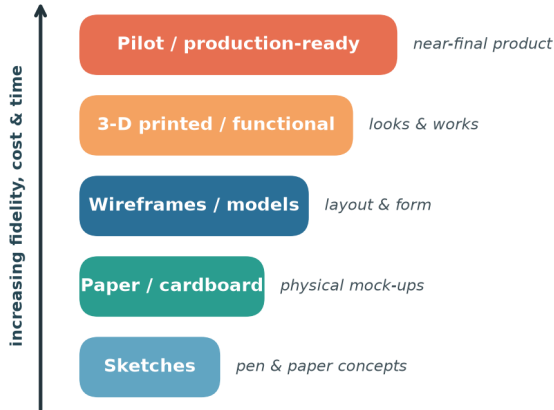
- ▶ Make ideas **tangible**.
- ▶ Test assumptions cheaply.
- ▶ Communicate with stakeholders.
- ▶ Fail fast, improve faster.

Types of prototype

- ▶ *Low-fidelity* — sketches, paper, cardboard.
- ▶ *Mid-fidelity* — wireframes, models.
- ▶ *High-fidelity* — 3-D printed, functional.

Prototype Fidelity Ladder

From cheap & rough to refined & functional



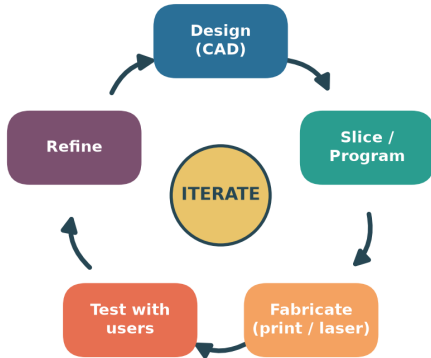
Start **low** and cheap, climb only as questions demand.

- ▶ Low fidelity answers *"is this the right idea?"*
- ▶ High fidelity answers *"does it work & feel right?"*

Each rung costs more time and money — so *learn cheaply first*.

Rapid Prototyping Cycle

Fail fast, learn faster — CAD to physical part in hours



Sample: laser-cut acrylic housing → test-group marketing → refine

Rapid prototyping uses digital fabrication (3-D printing, laser cutting, CNC) to go from **CAD** to a physical part in *hours*. **Testing & sample marketing**

- ▶ Test the prototype with a small **test group**.
- ▶ Gather reactions before scaling up.
- ▶ Refine, then repeat the loop.



IV

Design Thinking & Customer Centricity

Experience, feedback & re-creation

15 hours

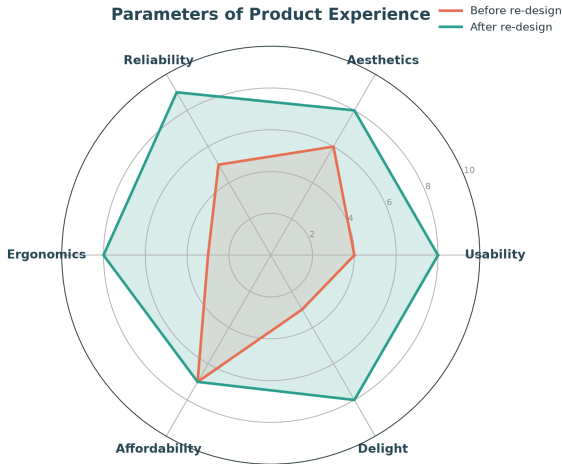
Real customers face everyday frustrations that design thinking can solve:

- ▶ Products that are **hard to assemble** or use.
- ▶ Confusing instructions & interfaces.
- ▶ Poor **ergonomics** causing discomfort.
- ▶ Long waits and clunky service.
- ▶ Features nobody asked for.
- ▶ A gap between the *promise* and the *experience*.

Design thinking to the rescue

By empathising with customers we uncover these pains and re-design the product experience around them.

Parameters of Product Experience



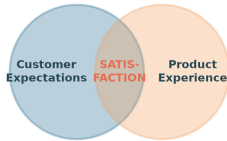
Product experience is multi-dimensional:

- Usability & Reliability
- Aesthetics & Ergonomics
- Affordability & Delight

Thoughtful re-design lifts every axis — turning a merely functional product into one customers *love*.

Aligning Customer Expectations

Aligning Customer Expectations with the Product



Gap too wide → disappointment · aligned → loyalty & advocacy

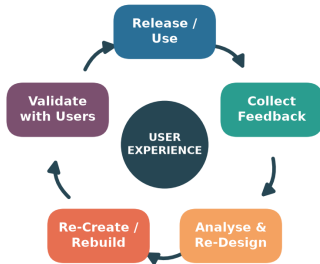
Satisfaction lives where *expectations* meet *experience*.

- ▶ Gap too wide → disappointment.
- ▶ Well aligned → loyalty & advocacy.

Manage expectations honestly *and* raise the experience to close the gap.

Feedback → Re-Design → Re-Create Loop

Continuous improvement centred on user experience



Address ergonomic challenges · close the gap between expectation and experience

The **feedback loop** keeps a product improving:

- 1 Release & use.
- 2 Collect feedback.
- 3 Analyse & re-design.
- 4 Re-create / rebuild.
- 5 Validate with users.

Focus relentlessly on **user experience** and address ergonomic challenges each cycle.

User-focused design

- ▶ Design *with* users, not just for them.
- ▶ Fit the product to the human — **ergonomics**.
- ▶ Test with real tasks in real contexts.

Toward the final product

- ▶ Converge feedback into refinements.
- ▶ Validate safety, comfort & usability.
- ▶ Ship — then keep listening.

Remember

A "final" product is really the *best current version* — the loop never truly ends.



Fabrication Laboratory

The route map: survey to service blueprint

15 hours

The Fab-Lab Route Map

Fab-Lab Route Map — From Survey to Service Blueprint



A step-by-step recipe for the lab project:

- 1 Conduct surveys.
- 2 Identify a problem.
- 3 Frame a problem statement.
- 4 Apply the **CREATE** tool.
- 5 Draw the product / system.
- 6 Build a Customer Journey Map.
- 7 Frame 2–3 **HMW** questions.
- 8 Design a Service Blueprint.

Conduct surveys

- ▶ Individually or as a group.
- ▶ Observe, interview, question.
- ▶ Look for **real, felt pains**.

Identify a problem worth solving.

Framing a problem statement

A good statement is **user-centred**, **clear** and **actionable**:

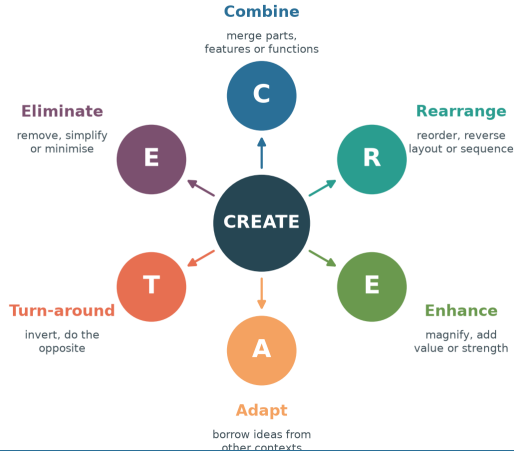
"[User] needs [need] because [insight]."

Avoid baking in a solution — describe the *problem*.

The CREATE Tool

The CREATE Ideation Tool

Six trigger words to transform an existing product



CREATE is a set of *trigger words* to transform an existing product:

- ▶ **Combine** **Rearrange**
- ▶ **Enhance** **Adapt**
- ▶ **Turn-around** **Eliminate**

Apply each trigger to your problem, then **draw the product / system** that results.

Customer Journey Map (CJM)

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Mapping the experience — before, during & after



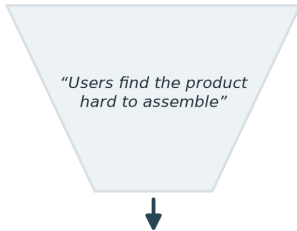
Touch-points feed the Service Blueprint · pain points become How-Might-We questions

Map actions, touch-points, emotions & pain points — before, during & after.

How Might We (HMW) Questions

Turning pain points into springboards for ideation

Observed pain point



How might we make assembly effortless?

How might we guide users step-by-step?

How might we remove assembly entirely?

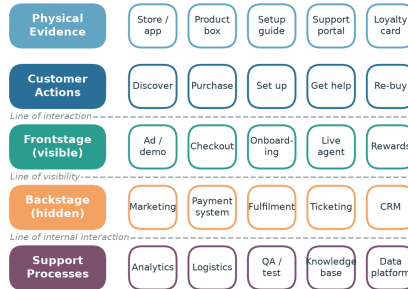
Frame 2–3 HMW questions — broad enough to inspire, narrow enough to act

Turn each **pain point** from the CJM into an optimistic **"How Might We..."** question.

- ▶ Broad enough to *inspire* ideas.
- ▶ Narrow enough to *act* on.
- ▶ Frame 2–3 for your mock scenario.

Service Blueprint

Layering the customer journey with what happens behind the scenes



Layer the journey with front-stage, back-stage & support processes; identify touch-points.

Bringing It All Together

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- ▶ **Unit I** — learn, remember, empathise.
 - ▶ **Unit II** — the design-thinking process.
 - ▶ **Unit III** — design & prototype products.
 - ▶ **Unit IV** — centre on the customer.
 - ▶ **Unit V** — build & blueprint in the lab.

The golden thread

Every unit returns to one idea: **empathise with people, then iterate** — observe, frame, ideate, prototype, test, and repeat until the experience truly serves the user.

Thank You

Design Thinking & Fabrication Laboratory

Empathise → Define → Ideate → Prototype → Test